



Project Description

The **Maternal Care Platform** is an AI-powered web application developed to support pregnant women by providing intelligent pregnancy monitoring, early risk assessment, and personalized healthcare guidance. The platform combines **Machine Learning**, **Optical Character Recognition (OCR)**, and an **AI Health Assistant** to deliver an integrated maternal healthcare experience.

The system enables users to register securely, enter their health information, and receive pregnancy risk predictions using a trained **Random Forest** machine learning model. In addition, users can upload laboratory reports in PDF or image format, where OCR technology automatically extracts medical values and analyzes them to generate risk predictions without requiring manual data entry.

Beyond risk prediction, the platform offers pregnancy tracking with weekly milestones, personalized health tips, an AI Health Assistant for pregnancy-related questions, and a history page that allows users to review previous predictions and medical reports.

The application is developed using **Python**, **Streamlit**, **SQLite**, **EasyOCR**, **OpenCV**, **Plotly**, and **bcrypt**, providing a secure, interactive, and user-friendly environment. By integrating multiple AI technologies into a single platform, the Maternal Care Platform aims to improve maternal health awareness, encourage early detection of pregnancy-related risks, and provide accessible healthcare support while emphasizing that all AI-generated recommendations should complement—not replace—professional medical advice.